

Human Resource Management System (HRMS)

MVP Scope & Project Overview

Project Type: Full Stack Web Application

Technology Stack: React.js, Express.js, MongoDB (Mongoose), JWT Authentication, Tailwind CSS

1. Project Overview

The Human Resource Management System (HRMS) is a web-based application designed to simplify and centralize essential HR operations within an organization. The MVP focuses on the core functionalities required for managing employees, attendance, leave requests, payroll, and employee self-service through a secure role-based system.

The application is intended to provide a clean, scalable architecture while demonstrating full-stack development practices including authentication, authorization, CRUD operations, relational data management, and reporting.

2. Project Objectives

- Centralize employee information.
 - Digitize attendance tracking.
 - Simplify leave management.
 - Generate monthly payroll records.
 - Provide employee self-service.
 - Implement secure authentication and role-based authorization.
 - Generate basic HR reports.
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3. User Roles

Admin

- Full access to the system.
- Manage employees.
- Manage departments.
- Manage attendance.
- Approve leave requests.

- Generate payroll.
- View reports.

HR

- Manage employees.
- Manage attendance.
- Approve leave requests.
- Generate payroll.
- View reports.

Employee

- Login securely.
 - View personal profile.
 - Mark attendance.
 - Apply for leave.
 - View attendance history.
 - View leave history.
 - View payroll history and payslips.
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4. MVP Features

4.1 Authentication & Authorization

Features

- Secure Login
 - JWT Authentication
 - Password Encryption
 - Role-Based Access Control
 - Protected Routes
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4.2 Dashboard

The dashboard provides an overview of HR statistics.

Widgets

- Total Employees
 - Employees Present Today
 - Employees on Leave
 - Monthly Payroll Summary
 - Recent Activities
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4.3 Employee Management

Features

- Add Employee
 - Edit Employee
 - Delete Employee
 - View Employee Details
 - Department Assignment
 - Designation Management
 - Employee Status (Active / Inactive)
 - Auto-generated Employee ID
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4.4 Department Management

Features

- Create Department
 - Update Department
 - Delete Department
 - View Department List
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4.5 Attendance Management

Features

- Daily Check-In
- Daily Check-Out
- Attendance History
- Monthly Attendance Records
- Attendance Status Tracking
- Working Hours Calculation

Attendance Status:

- Present
 - Absent
 - Half Day
 - Leave
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4.6 Leave Management

Features

- Apply Leave

- View Leave Requests
- Approve Leave
- Reject Leave
- Leave History
- Leave Remarks

Leave Types:

- Sick Leave
- Casual Leave
- Annual Leave
- Unpaid Leave

Leave Status:

- Pending
 - Approved
 - Rejected
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4.7 Payroll Management

Features

- Generate Monthly Payroll
- Store Salary Structure
- Payroll History
- Payroll Status
- Generate Payslip
- Download Payslip (PDF)

Payroll Information:

- Basic Salary
 - Allowances
 - Deductions
 - Net Salary
 - Payment Status
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4.8 Employee Self-Service

Features

- View Profile
- Update Personal Information

- Change Password
 - View Attendance
 - View Leave History
 - View Payroll History
 - Download Payslips
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4.9 Reports

Available Reports

- Employee Directory
- Attendance Report
- Leave Report
- Payroll Report

Reports can be exported for record keeping.

5. Database Models

The application consists of the following collections:

- Department
 - User
 - Attendance
 - Leave
 - Payroll
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6. Technology Stack

Frontend

- React.js
- React Router
- Axios
- Tailwind CSS

Backend

- Express.js
- Node.js

Database

- MongoDB
- Mongoose ODM

Authentication

- JWT
 - bcrypt
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7. Security Features

- JWT Authentication
 - Password Hashing
 - Protected API Routes
 - Role-Based Authorization
 - Input Validation
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8. Project Deliverables

- Responsive HRMS Web Application
 - REST API Backend
 - MongoDB Database Design
 - Authentication & Authorization
 - Employee Management
 - Attendance Management
 - Leave Management
 - Payroll Module
 - Employee Dashboard
 - Reporting Module
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9. Features Not Included in MVP

The following enterprise-level features are intentionally excluded from this MVP and can be implemented in future phases:

- Recruitment & Applicant Tracking System (ATS)
- Digital Onboarding & Offboarding Workflows
- Performance Management
- Learning & Development (LMS)
- Finance ERP Integration
- Biometric Device Integration
- Single Sign-On (SSO)

- Third-Party API Integrations
 - Email & Push Notifications
 - Advanced Analytics Dashboards
 - Audit Logs & Compliance Tracking
 - Multi-Company / Multi-Branch Support
 - Tax & Statutory Compliance Automation
 - Employee Document Repository
 - Shift & Roster Management
 - Travel & Expense Management
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10. Future Enhancements

Potential future improvements include:

- Mobile Application (Android/iOS)
 - Recruitment Module
 - Performance Appraisal System
 - Training & Learning Portal
 - Automated Payroll Calculations
 - Employee Document Uploads
 - Notification System
 - Advanced Analytics Dashboard
 - Biometric Attendance Integration
 - Cloud Deployment & Scalability
 - Multi-Organization Support
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Conclusion

This MVP delivers the core functionalities of a Human Resource Management System by providing secure user authentication, employee management, attendance tracking, leave management, payroll processing, and reporting. The architecture is designed to be modular and scalable, allowing additional enterprise features to be incorporated in future development phases while maintaining a clean and maintainable codebase.